AIR FROM VENTS NOT COLD ENOUGH [FULL-AUTO AIR CONDITIONER]

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DESCRIPTION	• Magnetic clutch operates but A/C system malfunctions
POSSIBLE CAUSE	**Note** • If the engine coolant temperature increases due to a cooling system malfunction, the fail-safe function disables the A/C operation. • Drive belt malfunction • Refrigerant pressure sensor malfunction • Blower unit or condenser malfunction • Cooling fan system malfunction • Condenser or related part malfunction • A/C unit or condenser malfunction • Receiver/drier or expansion valve malfunction (valve closes too much) • Refrigerant lines malfunction • A/C compressor system malfunction, insufficient compressor oil • Over filling of compressor oil, malfunction in expansion valve or A/C unit air mix link system • Evaporative temperature sensor malfunction

Diagnostic Procedure

Step	Inspection		Action
1	**CHECK MALFUNCTION SYMPTOMS** **Note** • Without i-stop, go to the next step. • Is malfunctions occur in only when operating the i-stop?	Yes	Perform the i-stop troubleshooting. (See FOREWORD [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See FOREWORD [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**INSPECT CLIMATE CONTROL UNIT FOR DTC** • Retrieve the climate control unit DTC using the M-MDS. (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) • Are there DTC displayed?	Yes	Go to the applicable DTC troubleshooting procedures. (See DTC TABLE [FULL-AUTO AIR CONDITIONER].)
		No	Go to the next step.
3	**INSPECT REFRIGERANT PRESSURE TO LOCATE MALFUNCTION** • Perform refrigerant pressure check. (See REFRIGERANT PRESSURE CHECK.) • Is the refrigerant pressure normal?	Yes	Go to the next step.
		No	Record the inspection result. • If the refrigerant high-pressure and low-pressure values are both high, go to Step 7. • If the refrigerant high-pressure and low-pressure values are approximately the same, go to Step 10. • If the refrigerant high-pressure and low-pressure values are both low, go to Step 12. • If there is a vacuum on the low pressure side and extremely low pressure on the high pressure side, go to Step 18. • If there is low pressure on the high pressure side and high pressure on the low pressure side, replace the A/C compressor, then go to Step 22. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • If the refrigerant pressure is other than above condition, go to Step 20.
4	**INSPECT REFRIGERANT SYSTEM PERFORMANCE** • Perform refrigerant system performance test. (See REFRIGERANT SYSTEM PERFORMANCE TEST.) • Is the operation normal?	Yes	Operation is normal. (Recheck malfunction symptoms.)
		No	Go to the next step.

Step	Inspection		Action
5	**INSPECT DRIVE BELT** • Inspect the drive belt. (See DRIVE BELT INSPECTION [SKYACTIV-D 1.5].) (See DRIVE BELT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is it normal?	Yes	Go to the next step.
		No	Adjust or replace the drive belt, then the next step. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6	**INSPECT REFRIGERANT PRESSURE SENSOR** • Inspect the refrigerant pressure sensor. (See REFRIGERANT PRESSURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is it normal?	Yes	Go to the next step.
		No	Repair or replace malfunctioning part according to inspection result, then go to Step 22.
7	**INSPECT COOLING FAN OPERATION** • Verify the cooling fan operation. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-D 1.5].) • Is the cooling fan operation normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning location according to the inspection results. Then go to Step 22.
8	**VISUALLY INSPECT CONDENSER** • Is the condenser fin clogged or obstructed by foreign material?	Yes	Remove the foreign material. Repair the condenser fin. Then go to Step 22.
		No	Go to the next step.
9	**CHECK REFRIGERATION SYSTEM FOR OVERCHARGE OR AIR CONTAMINATION** • Is the low side line hot to the touch?	Yes	Recover refrigerant. Evacuate system for one hour. Refill with correct amount of refrigerant, and go to step 22.
		No	Recover refrigerant, evacuate for **15 minutes**, refill with correct amount and go to step 22.
10	**CHECK TO SEE WHETHER MALFUNCTION IS IN EXPANSION VALVE OR ELSEWHERE** • Compare the refrigerant pressure of the low pressure side with the high pressure side at Step 3. • Is there little difference between the high pressure side and low pressure side readings (refer to graph in REFRIGERANT PRESSURE CHECK procedure)? (See REFRIGERANT PRESSURE CHECK.)	Yes	Replace the expansion valve. After performing the following servicing, go to Step 22. • Adjust the compressor oil to the specified level. • After discharging, charge with new refrigerant to the specified level.
		No	Go to the next step.
11	**INSPECT AIR MIX DOOR RELATED PART INSTALLATION** • Measure the voltages at the following climate control unit terminals when the temperature control dials are set to MAX COLD and MAX HOT. — Terminal G • Are voltages normal?	Yes	Adjust the compressor oil to the specified amount, then go to Step 22. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	• Inspect the air mix link, air mix crank, and air mix rod of the A/C unit correctly and securely installed to their positions. (See A/C UNIT REMOVAL/INSTALLATION.) • Repair or install correctly for suspect part according to inspection result, then go to Step 22.
12	**INSPECT BLOWER UNIT FOR BLOCKAGE** • Is the blower unit intake and air filter clogged? (See AIR FILTER INSPECTION.)	Yes	Remove the cause of the clogging. Replace the air filter if it is clogged. (See AIR FILTER REMOVAL/INSTALLATION.) Then go to Step 22.
		No	Go to the next step.
13	**CHECK TO SEE WHETHER MALFUNCTION IS REFRIGERANT LINE LEAKAGE OR ELSEWHERE** • Verify if there is gas leakage from the system hoses using the gas leak tester. • Is there gas leakage?	Yes	• If there is leakage from a system hose connection area, go to Step 15. • If there is leakage other than from a system hose connection area, go to Step 17.
		No	Go to the next step.

Step	Inspection		Action
14	**VISUALLY INSPECT REFRIGERANT LINE** • Is a system hose crushed?	Yes	Replace the crushed system hose. (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) After performing the following servicing, go to Step 22. • Adjust the compressor oil to the specified level. • After discharging, charge with new refrigerant to the specified level.
		No	Go to Step 21.
15	**CHECK TO SEE WHETHER MALFUNCTION IS IN REFRIGERANT LINE JOINT LOOSE OR O-RING** • Tighten the system hose connection area to the specified torque. (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) • Has the leakage stopped?	Yes	Go to the next step.
		No	Go to Step 17.
16	**VISUALLY INSPECT REFRIGERANT LINE** • Is a system hose crushed?	Yes	Replace the crushed system hose. (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) After performing the following servicing, go to Step 22. • Adjust the compressor oil to the specified level. • After discharging, charge with new refrigerant to the specified level.
		No	Adjust the compressor oil to the specified amount, then go to Step 22. (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See A/C COMPRESSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
17	**VISUALLY INSPECT REFRIGERANT LINE** • Is a system hose crushed?	Yes	Replace the O-ring of the leaking area. Replace the crushed system hose. (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See REFRIGERANT LINE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) After performing the following servicing, go to Step 22. • Adjust the compressor oil to the specified level. • After discharging, charge with new refrigerant to the specified level.
		No	Replace the O-ring of the leaking area. After performing the following servicing, go to Step 22. • Adjust the compressor oil to the specified level. • After discharging, charge with new refrigerant to the specified level.
18	**CHECK TO SEE WHETHER MALFUNCTION IS WATER IN REFRIGERANT SYSTEM OR ELSEWHERE** • Is there is no refrigerant pressure on the low pressure side, or is it normal? (See REFRIGERANT PRESSURE CHECK.)	Yes	Replace the condenser. (Water in refrigerant system) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 22.
		No	Go to the next step.

Step	Inspection		Action
19	**CHECK TO SEE WHETHER MALFUNCTION IS IN RECEIVER DRYER FILTER OR EXPANSION VALVE** • Remove the expansion valve and verify its condition. (See EXPANSION VALVE REMOVAL/ INSTALLATION.) • Is there refrigerant leakage or valve clogging?	Yes	**If there is foreign matter clogging the valve:** • Remove the foreign matter. **If there is refrigerant leakage or clogging:** • Replace the expansion valve. Perform discharge, charge with new refrigerant, and then go to Step 22.
		No	Replace the condenser. (Receiver/Dryer filter is clogged.) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See CONDENSER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Then go to Step 22.
20	**INSPECT EVAPORATIVE TEMPERATURE SENSOR** • Inspect the evaporator temperature sensor. (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is it normal?	Yes	Verify the evaporator temperature sensor position. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) Then go to Step 22.
		No	Replace the evaporator temperature sensor. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) Then go to Step 22.
21	**INSPECT AIR MIX DOOR RELATED PART INSTALLATION** • Measure the voltages at the following climate control unit terminals when the temperature control dials are set to MAX COLD and MAX HOT. — Terminal G • Are voltages normal?	Yes	Go to the next step.
		No	• Inspect the air mix link, air mix crank, and air mix rod of the A/C unit correctly and securely installed to their positions. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) • Repair or install correctly for suspect part according to inspection result, then go to next Step.
22	**VERIFY THAT MALFUNCTION SYMPTOM DOES OCCURS AFTER REPAIR** • If the refrigerant discharged during inspection has not been recharged, discharge and charge with new refrigerant to the specified level. • Does cool air blow out? (Are results of refrigerant system performance test normal?) (See REFRIGERANT SYSTEM PERFORMANCE TEST.)	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if malfunction recurs.